PHYSICS

Paper 5054/32
Practical Test

General comments

There was better discrimination in **Section A** of the paper compared to **Section B**. In **Question 4** many candidates scored well by recording scale readings in order to determine the depression of the rule and plotted a good graph of the results.

Comments on Specific Questions

Section A

Question 1

(a) The majority of candidates were awarded full credit for the measurement of temperature. The two main sources of error were:

- the omission of units or the wrong units for temperature, typically ° rather than °C,
- a temperature rise that was too small. It was expected that the temperature would rise by at least 5 °C but a small number of candidates obtained rises as low as 3 °C.

(b) A large number of candidates did not know that the heat gained by the water was measured in joules. The units of specific heat capacity were often quoted instead of joules. Credit was given for correct numerical calculations and units or lack of units was ignored. A number of candidates made errors in calculations because:

- average values were determined for the specific heat capacities,
- the specific heat capacity of the *metal* was used when considering the heat gained by the *water*,
- the specific heat capacity of *water* was used when considering the heat lost by the *metal*.

The temperature of the Bunsen flame should have been found by adding the final temperature of the water to the fall in temperature of the metal. Very few candidates used this procedure; most were awarded credit for **(i)** and **(ii)** only.

(c) Most candidates appreciated that heat was lost but it was not always clear that the heat was lost when the mass was transferred from the Bunsen flame to the water. A number of candidates talked about heat being lost from the mass when it was being heated, others talked about heat being lost from the metal to the water but neither of these points gained credit.

Question 2

Some candidates drew a normal from the point M and placed an object pin O, 10.0 cm from M, but did not know how to use the sighting pins to find the position of the image. Some other candidates correctly drew the normal and O and used sighting pins to locate the image, but did not use the best experimental techniques. All candidates would benefit from more practise in experimental techniques.

The more able candidates drew the normal and O and used sighting pins to locate the image but also used the following techniques:

(a) Placed the sighting pins along a particular ray a long way apart e.g. at least 5.0 cm apart.

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(b) Used rays on opposite sides of the normal and quoted the length of the line from I to the line XY to the nearest mm. Such candidates were likely to achieve full credit.

Question 3

(a) The switch, the capacitor and the voltmeter were in parallel with each other. The resistor and the power supply were then in series with this arrangement. Unfortunately a number of candidates drew the switch in series with the resistor and the power supply and this was not correct.

(b) The time taken to charge the capacitor to 2.0 V should have been in the region of 50 s. If the voltmeter used did not have a high resistance, then this time would have been longer. In the mark scheme, a time of up to 99 s was allowed. A number of candidates were misreading the stopwatch. An answer such as 1.10 s or even 1.10 minutes was more likely to be 70 s, i.e. 1 minute and 10 seconds. Since an average value for the time was required, it was expected that candidates would repeat their measurements but only the more able candidates did this.

(c) When the time for the capacitor to charge to 1.0 V was determined, the majority of candidates obtained the correct answer. Again, repeated measurements were expected.

(d) The ratio calculation was solely based on the candidates' measurements; no particular range of values was required. Many candidates were not awarded credit and there were 2 reasons for this:

- the ratio was given with units of seconds but it should have no units,
- the ratio was quoted to too many significant figures.

Section B

Question 4

(a) The set square should have been placed between the floor and the vertical metre rule in order to check that the rule was vertical. It was not always clear that one edge of the set square was in contact with the floor.

(b)(c) Many candidates gave correct answers to **(b)** and **(c)**. Most candidates showed the scale readings that enabled them to determine the two *y* values, measured those scale readings to the nearest mm and correctly recorded the mass values.

(d) Examiners expected stiff wooden metre rules to be used and so expected a maximum deflection somewhere in the region of 10 cm with the results producing a linear graph. In some cases Centres seemed to have used very flexible rules, possibly plastic, which gave deflections up to 50 cm and usually produced a graph that was a smooth curve. The mark scheme allowed for both types of graph to be produced and a large number of candidates obtained full credit for this section.

(e) Graph work was generally good. The most common error was in drawing the straight line or smooth curve. Lines should be thin and, if the graph is a straight line, candidates should use a 30 cm ruler rather than a 15 cm ruler to draw the line. This ensures that there are no kinks in the line where two short lines have been joined.

(f) Able candidates were awarded full credit for the gradient. Weaker candidates made the following mistakes:

- used 2 points on a curved graph to find the gradient - if the line is a curve, then a tangent should be drawn to the curve and the gradient of the tangent should be found,
- used a small triangle to find the gradient - the base of the triangle used to find the gradient should occupy at least half of the page in the appropriate direction,
- calculated a value to either 1 or more than 3 significant figures.

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